

A connected path for recyclable scrap

ScrapLink helps a household request a scrap pickup, helps a collector record the collection, and gives recyclers a shared place to offer for available material.

OUR IDEA

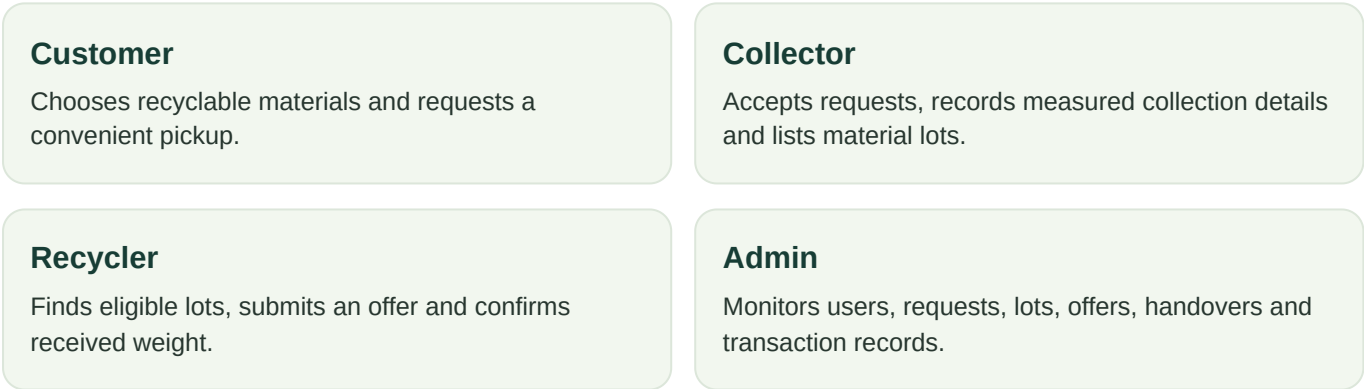
Keep the people and records in one workflow.

Pickup requests, collection weights, recycler offers and handover statuses are shown through role-based portals connected to one backend. Each person sees the information and actions relevant to their role.

The problem it addresses

When pickup coordination happens through separate conversations or handwritten notes, it can be hard to know whether a request was accepted, how much material was collected, or whether a recycler received it. ScrapLink provides shared IDs and saved statuses so these steps are easier to follow.

WHO USES IT



THE BASIC FLOW



The flow above summarizes features found in the active website and API.

How a pickup moves forward

The first part of the workflow turns a customer's request into a measured collection record.

1 Customer selects the material

The customer chooses one or more available materials, estimates the weight, and describes the condition. They enter an address, city and preferred date/time. The request can include notes and an eligible collector selection.

The API checks that the materials and values are valid, saves the request and returns a pickup ID. The request begins with a "requested" status.

2 Collector receives and accepts the request

Eligible collectors can be notified about new requests. A collector reviews the request and accepts or rejects it. When accepted, the request is assigned to that collector and moves into the in-transit stage. The customer's portal can show the updated status and assigned collector details.

3 Collector records what was collected

At the doorstep, the collector enters the actual measured weight, amount and payment method. The system saves the collection, updates the pickup timeline and creates a batch record. It also creates marketplace lots for supported material types, linked back to the original pickup.

The collector can also create a lot directly from the Collector portal, without a household pickup request.

What the customer can follow

The pickup ID, current status, collector assignment and saved timeline. The interface also provides QR and collection tracking views for supported records.

When internet is unavailable

Collector collection drafts and queued collection submissions can be kept in browser storage. The app retries queued work when online; a queued item becomes server-confirmed only after successful sync.

Simple explanation: the customer describes the scrap; the collector confirms the actual weight and completion. Estimated and measured amounts are separate pieces of information.

From material lot to handover record

After collection, the marketplace connects available material with recyclers that accept it.

4 Recycler finds a matching lot

The Recycler portal lists available lots for materials accepted by that facility. The recycler can filter by material or city, review weight and location, then submit an offer amount and optional note. An active, verified facility is required for marketplace offers.

5 Collector responds to the offer

The collector sees offers attached to their lots and can accept or decline a pending offer. When an offer is accepted, the lot is matched and other pending offers for that lot are closed. The recycler can then schedule a future pickup.

6 Recycler confirms receipt

After collection, the recycler records the received weight. ScrapLink saves a handover record and updates the related lot and batch status. Linked customer pickup timelines can also reflect that the material reached the recycler.

The recycler dashboard can record an external processing reference after receipt. That reference is saved as entered; ScrapLink does not independently check it with an outside authority.

7 Admin monitors shared activity

The Admin portal presents user and verification information, requests, lots, offers, handovers, transaction/payment statuses, saved price observations, charts and audit records. Admin can update collector/recycler verification and manage materials and price observations.

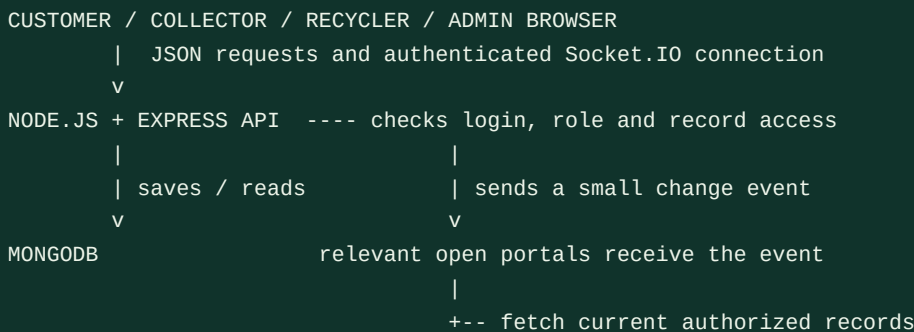
Payment records: what they mean

When the collector completes a customer pickup, a transaction and payment record are created. Cash is recorded as paid in the app. UPI and Direct Jan-Dhan selections remain pending because no digital payment provider is configured. A directly created lot can also create a pending payment record when the recycler confirms its handover.

Be clear with judges: ScrapLink records amounts and payment statuses; it does not currently move UPI or bank money. A recorded handover/reference is a workflow record, not independent proof that a physical recycling process occurred.

Shared records, updated portals

The website uses a browser app, a backend API, a MongoDB database and realtime notifications.



The frontend is built with React and TypeScript. Express exposes the API routes. MongoDB stores account and workflow data. Socket.IO sends a small event to relevant signed-in users when important data changes. The browser then fetches the latest records from the API, so other open portals update without a full page refresh.

Login returns a signed, time-limited JWT access token. The API checks the token and the user's role before allowing protected actions. Passwords are stored as hashes. Separate role portals help keep each person's screens focused on their part of the process.

Benefits of the idea

Clearer progress

Customers and collectors can refer to one request ID and its saved status instead of relying only on separate messages.

Measured records

Estimated weight from the booking can be compared with the actual weight entered at collection.

Recycler access

Material lots and offers create a shared way for collectors and eligible recyclers to coordinate a sale and handover.

Operational visibility

Admin can review saved requests, lots, offers, verification, handover and transaction information in one portal.

Example

A customer requests a paper pickup. A collector accepts it, weighs the paper and completes the collection. The paper lot appears for eligible recyclers. One recycler makes an offer; the collector accepts it. The recycler schedules pickup and confirms the received weight. Open portals fetch the saved changes, and Admin can monitor the record trail.

In one sentence: ScrapLink connects a household pickup request to collection, recycler offers and a recorded handover, while giving each role a shared view of progress.